

Ensuring AI Reliability in Education

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The 'Assurance of AI in Education' workshop, held on 25th April 2024 at Loughborough University, was organised by the Operational Research Society, Validate AI, and its Computer Science Department. This event focused on addressing the safe integration of AI in educational settings as part of a roadshow series. Each workshop in the series is dedicated to identifying and solving AI assurance problems specific to various sectors and development challenges. While most workshops have been held in the UK, some have taken place overseas.

The Loughborough workshop successfully brought together over 40 delegates from government, academia, and industry to discuss the necessity of robust assurance mechanisms for AI applications in education. The rapid advancement of AI technologies, particularly generative AI, offers significant potential to enhance educational practices. However, it also raises substantial challenges related to trust, data privacy, and ethical considerations. The workshop aimed to bridge these gaps by creating a collaborative environment where delegates could share insights, discuss best practices, and outline future steps. There was a heightened awareness of the need for risk mitigation, especially concerning the exposure of AI in education to minors and the importance of protecting their welfare.

Summary of the Day

The day commenced with a keynote talk on "Generative AI, Trust, and Integrity Issues" by Dr Firat Batmaz, the AI in Education Group leader from the Computer Science Department. Dr Batmaz highlighted the transformative potential of generative AI in education, emphasising applications such as creating personalised learning materials, automating administrative tasks, and enhancing student engagement. Additionally, the talk addressed challenges

associated with generative AI, including data privacy concerns and the necessity for robust assurance mechanisms to maintain trust.

Following the keynote, a panel discussion addressed key issues such as the evidence required by stakeholders to trust AI technologies, insights from previous AI implementations in education, the evolution of roles within educational institutions to optimise AI use, and strategies for engaging various stakeholders. The panel included Martin Hamilton, an Independent Digital Innovation Consultant; Aftab Hussain, the Information and Learning Technology Manager at Bolton College; Mark Kennedy, Co-Director of the Data Science Institute at Imperial College London; and James Bowsher-Murray, the Head of Systems, Reporting, and Quantitative Methods at OFSTED. A notable point of discussion was the comparison between the rapid spread of AI technologies and the diverse opinions about them, similar to the varied responses to COVID-19. The panel underscored that while AI can outperform humans in specific tasks, its integration must be approached thoughtfully and responsibly.

After the networking lunch, Dr Tirath Virdee, Director of Xeno AI, delivered a presentation titled "Data and AI in Education." His talk explored the transformative potential of AI and data in education. He emphasised the importance of data democratisation for accessible and quality data, while also highlighting challenges such as trust, security, and cost in AI adoption. Dr Virdee envisioned AI-driven changes including automated grading, improved learning pathways, and tutor bots, proposing a future where AI enhances personalised and efficient learning environments.

The afternoon continued with interactive sessions facilitated by Shakeel Khan from Validate AI. Participants were divided into six groups to discuss different aspects

of AI integration in education. Each group focused on one of the six AI assurance factors prescribed by Validate AI to evidence that AI is trustworthy, namely, how to manage scoping, data, algorithmic build, maintenance, as well as ethics and legal requirements. Topics covered included defining educational goals and conducting feasibility studies, exploring data sources and addressing privacy issues, designing and validating algorithms, performing performance testing and maintaining transparency, ensuring compliance with data protection laws, and addressing ethical considerations such as fairness and human oversight.

Key Issues Discussed

Several important issues were highlighted throughout the day, grouped into four main areas: trust and transparency, ethical considerations, technical readiness, and community engagement.

Trust and Transparency

Building trust in AI systems was emphasised as a priority, requiring transparent processes and clear communication with stakeholders. Robust validation methods are essential to ensure AI systems function as intended. Protecting student data and ensuring compliance with data protection laws are crucial for maintaining security and privacy.

Ethical Considerations

Ethical considerations were a major focus. Ensuring AI systems do not disadvantage any student groups is vital. Discussions emphasised the need for human oversight, accountability, and auditability. Transparent development and use of AI systems with clear ownership and regular audits to assess fairness and effectiveness are necessary. AI systems should promote equity and avoid biases that could target certain demographics. Establishing channels for reporting and correcting errors made by AI systems was also highlighted.

Technical Readiness

Technical readiness involved discussions on the iterative nature of algorithm design and criteria selection, stressing the importance of version control and stakeholder consultation. Both algorithm design and criteria selection should be part of an iterative process. Conducting performance testing and benchmarking AI against existing

educational practices is essential to validate its effectiveness. Setting up monitoring dashboards to track the AI's impact on educational outcomes over time provides actionable data for decisions and adjustments.

Community Engagement

Community engagement was seen as crucial for the successful adoption of AI technologies. This involves providing transparent explanations of the AI's role and impact to ensure understanding and buy-in from all stakeholders. Establishing feedback loops with educators and students to continuously improve AI applications is vital. Engaging a wide array of stakeholders, including parents and the local community, helps gain broader perspectives and support. Providing extensive training for educators on how to use AI tools and understanding their ethical implications and limitations is also necessary.

Future Steps in AI Assurance

The workshop concluded with a consensus on the next steps necessary to advance AI assurance in education. Developing comprehensive frameworks that encompass the technical, ethical, and legal aspects of AI assurance was highlighted as essential. Continuous monitoring systems and feedback loops with educators and students are crucial for the ongoing improvement of AI applications. Collaboration with policymakers is needed to develop regulations that ensure the safe and ethical use of AI in education. Additionally, providing training for educators and administrators to understand AI technologies and their applications in educational settings is vital. These steps are seen as necessary to build trust and enhance the effectiveness of AI technologies in education.

A summary of the workshop and a first draft framework for AI in Education will be shared in the coming weeks and help from the wider community will be sought to build on the foundations set by the Loughborough workshop. 